

## Markscheme

Mathematics: analysis and approaches

Practice question bank

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44. (a)

Consider the equation  $\log_x 27 = \frac{3}{2}$ .

Find the value of  $x$ .

rewrite in exponential form:  $x^{\frac{3}{2}} = 27$  (M1)

$$x = 27^{\frac{2}{3}} = \left(27^{\frac{1}{3}}\right)^2$$
(M1)

$$x = 9$$
A1

[6 marks]

(b)

Hence find the value of  $\log_3 x$ .

$$\log_3(9)$$
(M1)

$$= 2$$
A1

[2 marks]

(c)

Hence find the value of  $\log_x 3$ .

$$\log_x 3 = \frac{1}{\log_3 x}$$
(M1)

$$= \frac{1}{2} = 0.5$$
A1

[2 marks]